Rissi Kumar Prabhakaran

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https://rissikumarp.github.io/rissikumar_portfolio

Education

University of Colorado Boulder — Boulder, CO
Master of Science in Data Science; GPA: 3.7/4.0 — *Aug 2024 - May 2026*

Anna University — Chennai, India
Bachelor of Technology in Artificial Intelligence and Machine Learning; GPA: 3.4/4.0 — *June 2020 - May 2024*

Experience

Kashmir World Foundation [Link] — May 2025 - May 2026
AI Data Scientist Intern — *Great Falls, Virginia*

- Deployed AI pipelines using **BirdNET** and **YamNET** for bioacoustic data preprocessing, processing **100,000+ audio samples** through segmentation and species analysis with spectrogram integration, improving model precision by **11%**.
- Developed an Autonomous Vehicle Assistant, a multi-agent system for drone operations utilizing **Agno framework** for agent orchestration, real-time navigation, telemetry monitoring, and multimodal data analysis across **3+ coordinated agent workflows**, generating structured status reports for mission decision support.
- Orchestrated end-to-end drone telemetry pipelines connecting **ArduPilot SITL** to LLM-based agents via **DroneKit**, **MAVLink protocol**, and TCP relay systems.
- Architected multi-agent architectures with **OpenRouter API**, reducing inference cost by **21%**, implementing context engineering and designing **RAG knowledge systems** with vector DB for agent context retrieval.

Classic Engineering Corporation [Link] — June 2022 - July 2024
AI Engineer Intern — *Chennai, India*

- Reduced VMC line scrap rate, eliminating expenditure, by designing and deploying a real-time defect detection system using **YOLOv5x**, served via **FastAPI** and containerized with **Docker** on a **Jetson Nano** edge device at the machine exit station.
- Achieved 94.2% defect classification accuracy and cut a client's incoming QC rejection rate by building a controlled ring-lit inspection pipeline with a fixed industrial camera, and augmentation using **OpenCV** and **Albumentations**, with real-time inference delivering sub-1.2 second verdicts tracked against downstream rejection records via **MLflow**.
- Flagged high-risk hydraulic cylinders before dispatch by building a predictive failure classifier using **TSFresh** automated feature extraction and **XGBoost**, trained on 3 years of pressure decay time series cross-referenced against field warranty claim records.
- Cut per-query search time for the engineering team by building and deploying an internal document Q&A tool over 12 years of maintenance logs and fixture failure reports using **LangChain**, **FAISS** vector indexing, and the **OpenAI API.**

Vlog Innovations [Link] — January 2023 - March 2023
AI Engineer Intern — *Chennai, India*

- Engineered a document ingestion and vector indexing pipeline using **FAISS** and **sentence-transformer embedding models**, processing and embedding internal training documents through batch transformation scripts for an internal knowledge search prototype.
- Evaluated semantic search quality across multiple retrieval configurations, benchmarking **precision@k and recall metrics** using **Python (Pandas, NumPy)** across training materials to optimize retrieval relevance for downstream **GPT-3.5** query answering and delivering optimization recommendations.
- Designed and executed content moderation classification experiments comparing **DistilBERT** against **Logistic Regression** and **SVM** models, building pipelines and orchestrating experiment metadata with **MLflow**, tracking precision-recall tradeoffs and hyperparameter sensitivity across **50+ runs** to identify optimal tagging thresholds.

Pantech E-Learning [Link] — June 2022 - June 2022
AI Engineer Intern — *Chennai, India*

- Built a content recommendation system using **RoBERTa** with **HuggingFace Transformers** and **PyTorch** for semantic course matching, analyzing semantic similarity patterns across **15,000+ learning resources** and computing similarity scores with **MySQL**.

Technical Skills

Languages: Python, R, SQL, C, C++, JavaScript, HTML, CSS, Bash, MATLAB, Scala, Julia, PySpark, Java

ML/DL & NLP: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face, Transformers, LLMs, BERT, GPT, RAG, OpenCV, NLTK, SpaCy

Agentic AI: GPT-4, Claude, Gemini, Fine-tuning, Prompt Engineering, Multi-Agent Systems, LangChain, LlamaIndex, Agno AI, RAG Systems, Context Engineering

Data Science & Statistics: Pandas, NumPy, SciPy, Matplotlib, Seaborn, A/B Testing, Hypothesis Testing, Time Series Analysis, Feature Engineering, KPI Development, Regression, Predictive Modeling, Cohort Analysis, Funnel Analysis

MLOps & Big Data: Docker, Kubernetes, MLflow, Apache Spark, Apache Kafka, Hadoop, PySpark, Databricks, Apache Airflow, Prefect, dbt, HDFS, Hive, CI/CD

Cloud & Databases: AWS (SageMaker, S3, EC2, Glue, Lambda, Redshift, EMR), GCP (Vertex AI, BigQuery, Dataflow), Azure (ML, Data Factory, Synapse), PostgreSQL, MySQL, MongoDB, Snowflake, Redshift, BigQuery, Redis, Pinecone, ChromaDB, FAISS

BI & Visualization: Power BI, Tableau, Excel (Pivot Tables, VBA, Power Query), Looker, Google Analytics, Plotly, Dash

Data Modeling & APIs: Star Schema, Snowflake Schema, Dimensional Modeling, Data Normalization, RESTful APIs, GraphQL, Flask, FastAPI, Microservices

Projects

Synthetic Data Generation Platform With Hallucination Detection [Link]

- Structured a multi-agent validation system with an orchestrator agent coordinating **4 specialized agents** (rule-based, statistical, LLM semantic, RAG similarity) featuring inter-agent communication, autonomous decision-making, and self-healing pipeline capabilities including automatic **CTGAN re-generation** on quality threshold failures.
- Constructed hierarchical hallucination detection system using fine-tuned GPT-4o-mini (trained on 1,000 domain-specific fraud examples via **OpenAI Fine-Tuning API**), achieving 98% validation accuracy with 12% improvement over base model across CTGAN output validation and LLM reasoning verification through meta-validation layer.
- Integrated **RAG-based similarity validation** using **Pinecone vector DB** and **OpenAI text-embedding-3-small** for nearest-neighbor anomaly detection with automated escalation to LLM semantic agent, deployed via **FastAPI REST API** and **Docker**.

Retrieval Based Question Answering Model for Medical Drugs [Link]

- Built a RAG-based medical drug QA system using MiniLM-V6, FAISS, and Sentence-BioBERT reranking across 45,697 chunks from 2,755 drugs, benchmarked 4 embedding models, and integrated Llama-4, achieving 87.50% accuracy and 60.96% F1@3.

AI-Powered Big Data Observability Platform [Link]

- Engineered a **RAG pipeline** using **sentence-transformers** with **ChromaDB** across 4 isolated collections with multi-collection similarity search and distance-based ranking, paired with structured JSON LLM output via **Ollama** with root cause, remediation steps, and confidence scoring across Infrastructure/CodeLogic/DataQuality failure categories.
- Architected a pipeline observability chatbot with **FastAPI** and **React (Vite + TailwindCSS)**, dynamically assembling LLM prompts from **ChromaDB** retrieved code chunks, live **Airflow** task logs via REST API, **Kubernetes** pod logs via the Python kubernetes client, and **Marquez OpenLineage** lineage events, enabling natural language root cause diagnosis across multi-DAG pipeline failures.
- Built an **Airflow** pipeline engineering features from curated data, training a **RandomForestClassifier** with artifact serialization, containerized with **Docker** and deployable to **Kubernetes (EKS/GKE/AKS)** with PVC-backed data lake storage.

Smart Fields - Enhancing Agriculture with Machine Leanring [Link] *PyTorch, ResNet, Random Forest, Flask*

- Tackled crop yield prediction challenges by building **LSTM networks** with **weather data integration** (temperature, rainfall, humidity), achieving **90% accuracy** via **multi-variable correlation analysis** on **15+ environmental parameters**.
- Engineered **CNN-based disease detection** system using **ResNet-9** architecture across **38 disease categories** from **10,000+ leaf images**, achieving **94% F1-score** with **k-fold cross-validation**.
- Created **Random Forest** crop recommendation engine processing **7 soil parameters** with **real-time weather API** integration for data-driven farming decisions.

Task Insights and Workload Overview Analysis for Denver International Airport [Link]

- Built an integrated **Power BI dashboard** connecting **ServiceNow** enhancement requests with **Azure DevOps** tasks for Denver International Airport, providing real-time visibility into **53 active requests** across **27 customer stakeholders**.
- Designed automated priority-based timeline calculations, reducing manual tracking time by **15 minutes per analysis**, and implemented **NLP keyword extraction** for request summarization with calculated fields for cross-team dependency tracking, workload distribution, and monthly trend analysis.

Fire in Focus: A Deep Learning Approach to Analyzing Wildfires [Link]

- Assessed **25 years of wildfire patterns** across the United States using **NASA satellite imagery**, collecting and cleaning **1 million temporal records** across **10 Southern California counties** and performing EDA using **Python**.
- Built predictive classification models combining satellite hotspot detection with **40+ weather variables** to forecast wildfire probability, applying **Random Forest** and **XGBoost**, achieving **83.5% accuracy** and **0.87 AUC-ROC** with optimized KNN.

Spatiotemporal Deep Learning for High-Resolution Flood Mapping [Link]

- Built a data pipeline to extract, transform, and analyze **NASA satellite** and geospatial flood records for Houston's urban zones, processing raw **GeoTIFF rasters** into structured tabular datasets with spatial coordinates, timestamps, and flood-extent labels.
- Conducted comparative statistical analysis across **4 modeling approaches**, evaluating performance through precision, recall, and overlap metrics **(IoU)**, identifying that post-processing spatial filters improved boundary accuracy by **12%**.
- Tuned model hyperparameters using **Bayesian optimization (Optuna)**, reducing validation loss by **9.6%** over baseline.

Publications

Smart Fields: Enhancing Agriculture with Machine Learning [Link] – IEEE AIMLA 2024